

Why Mean Regression Fails Behavior Cloning: A Unified Gradient- and Representation-Level Account of the Regression–MIP Gap

Theory note draft

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Abstract

We give a unified account of why L_2 behavior-cloning regression underperforms two-view denoising objectives (MIP) on both human-demonstrated and scripted manipulation data, from the joint viewpoint of *gradient dynamics* and *representation dynamics*. The organizing object is the *realizability* of the training target given the network input. For state-only regression the target contains an unrealizable component — cross-demonstration disagreement on human data, near-degenerate label micro-differences on scripted data — and the two observed failure modes are the two ways this component can damage training: (I) when it is heavy-tailed at macroscopic scale, it captures the batch gradient of the shared encoder and destroys previously learned structure (optimization pathology); (II) when it is microscopic and degenerate, it withdraws the incentive to keep distinct states distinct, and the encoder folds (representation pathology). The anchor view of MIP makes every training target realizable, which simultaneously removes the noise gradient of (I) and creates a distinction incentive of relative strength $(\delta/\sigma)^2$ against (II). On the regression side, (I) is replaceable by a bounded-influence likelihood with learned state-conditional scale (Student- t), and the conditioning deficit is replaceable by observation history; (II) has no known loss-side replacement. Every proposition is paired with a measured quantity from our experiments. The note does not claim distributional multimodality is the operative difficulty; the measured disagreement is timing jitter around a single strategy, and the analysis proceeds entirely at the level of training signals.

1 Setup

Let $s \in \mathcal{S} \subset \mathbb{R}^{d_s}$ be an observation (possibly a window of past frames) and $a \in \mathbb{R}^{d_a}$ an action chunk. Demonstrations supply pairs (s_i, a_i) with

$$a_i = \mu(s_i) + \eta_i, \quad \mathbb{E}[\eta \mid s] = 0, \quad (1)$$

where μ is the state-conditional mean and η collects the component of the action that is *not a function of the input*: across demonstrations, operators at indistinguishable states act at different times, speeds, and grasp instants. The policy is $f_\theta = g_\theta \circ \phi_\theta$ with a shared encoder ϕ_θ and action head g_θ ; both views of the analysis concern the gradient received by the shared parameters of ϕ_θ .

Definition 1 (Realizable component). *Given a network input x (which may be s , an observation window, or an augmented input), the target a decomposes as $a = m^*(x) + \eta(x)$ with $m^*(x) =$*

$\mathbb{E}[a \mid x]$. We call $\eta(x)$ the unrealizable component relative to x : no measurable function of x can reduce it.

Two empirically measured data regimes anchor the analysis.

Assumption 1 (Human regime). $\eta \mid s$ is heavy-tailed with tail index $\nu \approx 2$ and strongly state-dependent scale: $\text{Var}(\eta \mid s)$ varies by orders of magnitude across task phases, and the top-decile states by conditional scale carry a disagreement dominated by discrete timing events (gripper open/close instants, pause-versus-move). Measured: tail energy of the top decile is 85% gripper channel; kurtosis of the per-sample deviation ≈ 10.6 (tool-hang), 8.1 (square-mh).

Assumption 2 (Scripted regime). $\eta \approx 0$, but the labels are degenerate: there exist large state regions R on which labels differ only microscopically, $a(s) = \bar{a} + \delta u(s)$ with $\|\delta u\| \ll$ the typical action scale, while closed-loop correctness requires distinguishing states inside R .

2 Gradient dynamics under L_2 (human regime)

The per-sample loss $\ell_i = \|f_\theta(s_i) - a_i\|^2$ contributes to the encoder the gradient

$$g_i = -2 J_i^\top r_i, \quad J_i = \frac{\partial f_\theta}{\partial \phi_\theta}(s_i), \quad r_i = a_i - f_\theta(s_i). \quad (2)$$

Once the realizable part is fit, $r_i \rightarrow \eta_i$: the residual, and hence the per-sample encoder gradient, is the unrealizable component itself.

Proposition 1 (Gradient capture and churn). Under Assumption 1 with tail index $\nu \leq 2$, $\mathbb{E}\|\eta\|^2 = \infty$ (Student- t boundary case), and for a batch B the noise part of the batch gradient $\sum_{i \in B} J_i^\top \eta_i$ is dominated by its top order statistics: the ratio of the largest summand to the sum does not vanish as $|B| \rightarrow \infty$ (regular variation), and the signal-to-noise ratio of the batch mean gradient tends to zero. Consequently the shared encoder performs a heavy-tailed random walk driven by irreducible residuals, which degrades components of ϕ_θ that carry already-learned structure.

Derivation sketch. For regularly varying $\|\eta\|$ with index $\nu \leq 2$, sums and maxima are of the same order (single-big-jump principle); the empirical mean does not concentrate. Since $\eta \perp$ the learnable signal, the noise gradient is an unbiased zero-mean perturbation of unbounded variance applied to *shared* parameters, i.e. a random reparameterization of features used elsewhere. $\square$

Measured counterparts. Top-10% of samples carry 74%–77% of the encoder gradient under L_2 ; the influence distribution’s kurtosis grows $36 \rightarrow 4041$ over training (spike churn); behavioral acquire-then-unlearn (partial insertion servo at 37k steps, lost by 300k, SR $43 \rightarrow 31$); fit on the SR-critical low-disagreement samples is up to $9\times$ worse than under the robust loss at matched steps.

Proposition 2 (Bounded influence restores the signal). Replace ℓ_i by the Student- t negative log-likelihood with state-conditional scale $\sigma(s)$, $\rho(r) = \frac{\nu+1}{2} \log(1 + \|r\|^2/(\nu\sigma^2 D))$. Its influence function $\psi(r) = (\nu+1)r/(\nu\sigma^2 + \|r\|^2/D)$ is bounded, so the noise-gradient has finite variance and the batch SNR is restored; and with $\sigma(s)$ fit by maximum likelihood the per-sample weight is asymptotically inverse to the conditional scale — the efficient reweighting under the assumed family. The predicted influence ratio between top-decile disagreement samples and typical samples is $\psi(\eta_{\text{tail}})/\psi(\eta_{\text{typ}}) < 1$ once $\sigma(s)$ calibrates.

Measured counterparts. Own-view top-1% share 5–9% late (versus 59% for L_2), kurtosis 9.5–23; tail/typical influence 0.60; the calibration transient is visible (≈ 50 k steps of L_2 -like behavior before σ fits); SR 31–43 $\rightarrow$ 75–88. Hard removal of the same samples (trimming, kNN smoothing) *hurts* — the information is bounded, not deleted — consistent with the estimator view and inconsistent with a “noise filtering” view.

Proposition 3 (The anchor removes the unrealizable component). *The MIP anchor view supplies input $x = (s, a + \sigma\varepsilon, t^*)$ with target a . Then a is realizable given x up to the Bayes error of denoising, which the identity-plus-correction map attains; residuals vanish for every sample, including the disagreement tail. Hence the anchor view contributes no irreducible noise gradient, and no reweighting is needed: $g(\text{tail})/g(\text{typ}) \approx 1$.*

Measured counterparts. MIP own-view tail ratios 0.85–1.04 and top-1% shares 3–17% throughout training; evaluating the *anchor-free* readout of an anchor-trained network shows residuals concentrated precisely on the tail (ratios 3.9–4.1) — the network never learned those targets from the state; the anchor carried them.

Remark 1 (Conditioning as an alternative route to realizability). *Realizability can also be improved on the input side: if part of η is predictable from longer observation history h , then $\text{Var}(\eta \mid s, h) < \text{Var}(\eta \mid s)$ and the unrealizable component shrinks before the loss ever acts. Measured: extending the window from 2 to 8 frames cuts the tail-sample fitting error 4–5 $\times$ for both losses early in training, while leaving the learnable-majority fit nearly unchanged; the loss and history factors act on complementary terms, and success rates combine superadditively (12/28/35/60 across the 2×2).*

Remark 2 (Estimator location). *Where $\eta \mid s$ is a mixture over timing (moving versus paused), the conditional mean is itself a biased action: on transport its speed is $\approx 0.5\times$ the demonstrated speed in closed loop. No reweighting moves a location estimate off the mean; a quantile relocation along the demonstrated direction (pinball loss on the parallel speed component, $\tau > 1/2$) does, and only there. This is a second, logically independent defect of the mean.*

3 Representation dynamics under label degeneracy (scripted regime)

Proposition 4 (No distinction incentive). *Under Assumption 2, let ϕ_θ be collapsed on R ($\phi_\theta(s) = \bar{\phi}$ for $s \in R$). The excess L_2 risk of the collapsed encoder is $O(\|\delta\|^2)$, and the loss gradient toward separating R ’s states scales as $O(\|\delta\|^2)$ as well — for microscopic δ it is dominated by any competing compression pressure (weight decay, simplicity bias). The encoder therefore folds: states requiring different downstream behavior share a representation cell. The train loss cost is $O(\|\delta\|^2)$, but the closed-loop cost is $O(1)$: at deployment, neighborhoods are misassigned and the policy reads out the wrong region’s behavior.*

Measured counterparts. Distinction sensitivity of L_2 and Cauchy on scripted data 0.95–0.98 (blind); settle-phase embedding neighborhoods 4% pure; deployed failure is an attenuated readout of wrong-phase actions (cosine +0.97 to the wrong region’s labels at half amplitude). Note the loss-side lever of Section 2 is inert here: robust reweighting acts on residual *scale*, and all residuals are microscopic.

Proposition 5 (The anchor creates a distinction incentive of relative strength $(\delta/\sigma)^2$). *Consider two states $s_1, s_2 \in R$ with labels $\bar{a} \pm \delta/2$ and the anchor view with noise level $\sigma \gg \|\delta\|$. If ϕ_θ is collapsed, the denoiser must estimate the label from the anchor alone; the two-point mixture posterior yields excess denoising risk*

$$\Delta = \frac{\|\delta\|^2}{4} \left(1 - \mathbb{E} \tanh^2 \left(\frac{\delta^\top (\sigma \varepsilon)}{2\sigma^2} + \frac{\|\delta\|^2}{4\sigma^2} \right) \right) \approx \frac{\|\delta\|^2}{4} \left(1 - O\left(\frac{\|\delta\|^2}{\sigma^2}\right) \right), \quad (3)$$

which a separated encoder reduces to zero. Normalizing by the view’s overall loss scale σ^2 , the relative incentive to separate is $\Theta((\|\delta\|/\sigma)^2)$ per unit of anchor-view weight — whereas for L_2 the relative incentive is $\Theta(\|\delta\|^2)$ against an $O(1)$ loss scale. The anchor thus amplifies microscopic label distinctions into a first-order representation pressure, with an explicit dose control σ .

Measured counterparts. Distinction sensitivity of the anchor view 13× that of L_2 on scripted data, with measured dose–response $(\delta/\sigma)^2$; smaller σ increases the incentive (until anchors under-mix). We know of no loss reweighting that reproduces this: any $\rho(\|r\|)$ evaluated at $\|r\| = O(\delta)$ gives an incentive vanishing with δ .

4 The unified statement

	Human regime	Scripted regime
unrealizable component	macroscopic, heavy-tailed	microscopic, degenerate
pathology	gradient capture, churn (Prop. 1)	representation fold (Prop. 4)
level	optimization	representation
MIP’s fix	realizability (Prop. 3)	incentive $(\delta/\sigma)^2$ (Prop. 5)
regression-side fix	Student- t + history (+ quantile)	<i>open</i>
measured signature	top-share, kurtosis, tail ratio	sensitivity, neighborhood purity

Both regimes instantiate one principle: *L_2 regression charges the network for the component of the target it cannot realize, and the damage is determined by the scale distribution of that component.* Heavy macroscopic scale damages optimization; degenerate microscopic scale damages representation. The anchor view is a single mechanism that neutralizes both (realizability removes the charge; the anchor-conditioned Bayes problem re-prices microscopic distinctions at $(\delta/\sigma)^2$). On the regression side the first regime is fully replaceable by matched training signals; the second is not, and this asymmetry is exactly where a denoising-class objective is irreplaceable.

5 Relation to Jacobian-regularization accounts

A complementary hypothesis holds that denoising objectives implicitly regularize the conditioning Jacobian of the recovered clean map (schedule coefficient κ_t). Two of our measurements bound its role in the present setting. First, for the x_0 -parameterization used by MIP, $\kappa_t = 1$ and no shrinkage is implied; consistently, measured boundary effective-Lipschitz constants of MIP and regression policies coincide (2.6–3.7) while their success rates differ by 40 points. Second, enforcing output flatness directly (finite-difference penalties at the deficit radius, normal-direction jitter, shell-derivative penalties) degrades or destroys the policy at every dose tried, while the successful interventions leave the Jacobian unchanged. In this domain the operative channels

of the denoising signal are the two identified above — gradient realizability and representation incentive — not clean-map smoothness. The Jacobian account may govern other regimes (e.g. genuinely low-variance conditionals with $\kappa_t < 1$ parameterizations); the two accounts are logically independent and empirically separable by the probes used here.

6 Design implications

1. **Diagnose the unrealizable component first:** its tail index, alignment with a data-defined minority, channel decomposition, and history-resolvability determine which fix applies.
2. **Human-style data:** bounded-influence likelihood with learned state-conditional scale (Student- t , ν set to the measured tail index); observation history to the resolvability plateau; quantile relocation only on coordinates where the mean is demonstrably biased.
3. **Scripted-style data:** no loss reweighting suffices; a distinction incentive is required (anchor view with σ chosen by the $(\delta/\sigma)^2$ trade-off, or data-side diversification).
4. **What not to do:** hard sample removal (deletes bounded information), isotropic input noise with fixed labels (destroys closed-loop gain), output-flatness penalties (attack magnitude, not direction), inverse-residual channel weighting (re-concentrates on noise floors).

7 Limitations

The propositions are stated at the level of estimator and incentive structure, not end-to-end learning dynamics; the churn mechanism in Proposition 1 is supported by measured stream statistics and behavioral unlearning rather than a convergence theorem. The $(\delta/\sigma)^2$ computation treats a two-point mixture with a collapsed encoder; the constant in front depends on architecture. Success-rate arithmetic (which points a fix recovers) is task-dependent and is reported empirically in the companion experimental tables.